

Online Appendix for
Regional Heterogeneity and the Refinancing Channel of
Monetary Policy

Martin Beraja, Andreas Fuster, Erik Hurst and Joseph Vavra

August 18, 2018

A.1 MSA GROUPS USED IN FIGURES III, IV, AND V

Note: For large MSAs that are subdivided into Metropolitan Divisions, we use the latter (throughout the paper).

MSAs in the quartile with lowest E^{med} in November 2008:

Akron, OH; Anderson, IN; Bakersfield, CA; Bangor, ME; Battle Creek, MI; Bay City, MI; Bradenton-Sarasota-Venice, FL; Canton-Massillon, OH; Cape Coral-Fort Myers, FL; Carson City, NV; Cleveland-Elyria-Mentor, OH; Dalton, GA; Danville, IL; Dayton, OH; Deltona-Daytona Beach-Ormond Beach, FL; Detroit-Livonia-Dearborn, MI; El Centro, CA; Elizabethtown, KY; Elkhart-Goshen, IN; Fairbanks, AK; Flint, MI; Fort Lauderdale-Pompano Beach-Deerfield Beach, FL; Fort Walton Beach-Crestview-Destin, FL; Fort Wayne, IN; Fresno, CA; Grand Rapids-Wyoming, MI; Greeley, CO; Hagerstown-Martinsburg, MD-WV; Hanford-Corcoran, CA; Holland-Grand Haven, MI; Indianapolis-Carmel, IN; Jackson, MI; Jacksonville, FL; Kalamazoo-Portage, MI; Kankakee-Bradley, IL; Lake Havasu City-Kingman, AZ; Lakeland-Winter Haven, FL; Lansing-East Lansing, MI; Las Vegas-Paradise, NV; Madera-Chowchilla, CA; Mansfield, OH; Memphis, TN-MS-AR; Merced, CA; Miami-Miami Beach-Kendall, FL; Modesto, CA; Monroe, MI; Muskegon-Norton Shores, MI; Napa, CA; Naples-Marco Island, FL; Niles-Benton Harbor, MI; Oakland-Fremont-Hayward, CA; Ocala, FL; Orlando-Kissimmee, FL; Oxnard-Thousand Oaks-Ventura, CA; Palm Bay-Melbourne-Titusville, FL; Palm Coast, FL; Panama City-Lynn Haven-Panama City Beach, FL; Pensacola-Ferry Pass-Brent, FL; Phoenix-Mesa-Scottsdale, AZ; Port St. Lucie, FL; Providence-New Bedford-Fall River, RI-MA; Punta Gorda, FL; Redding, CA; Reno-Sparks, NV; Riverside-San Bernardino-Ontario, CA; Sacramento-Arden-Arcade-Roseville, CA; Saginaw-Saginaw Township North, MI; Salinas, CA; San Diego-Carlsbad-San Marcos, CA; Santa Rosa-Petaluma, CA; Sebastian-Vero Beach, FL; Springfield, OH; St. George, UT; Stockton, CA; Sumter, SC; Tampa-St. Petersburg-Clearwater, FL; Terre Haute, IN; Toledo, OH; Vallejo-Fairfield, CA; Visalia-Porterville, CA; Warren-Troy-Farmington Hills, MI; Weirton-Steubenville, WV-OH; West Palm Beach-Boca Raton-Boynton Beach, FL; Wheeling, WV-OH; Winchester, VA-WV; Worcester, MA; Youngstown-Warren-Boardman, OH-PA; Yuba City, CA; Yuma, AZ.

MSAs in the quartile with highest E^{med} in November 2008:

Albany-Schenectady-Troy, NY; Alexandria, LA; Anderson, SC; Asheville, NC; Athens-Clarke County, GA; Austin-Round Rock, TX; Baltimore-Towson, MD; Barnstable Town, MA; Baton Rouge, LA; Beaumont-Port Arthur, TX; Bellingham, WA; Bethesda-Frederick-Rockville, MD; Billings, MT; Binghamton, NY; Bismarck, ND; Blacksburg-Christiansburg-Radford, VA; Boulder, CO; Bridgeport-Stamford-Norwalk, CT; Buffalo-Niagara Falls, NY; Burlington-South Burlington, VT; Cambridge-Newton-Framingham, MA; Cedar Rapids, IA; Charleston, WV; Charlottesville, VA; Cleveland, TN; College Station-Bryan, TX; Corvallis, OR; Cumberland, MD-WV; Dubuque, IA; Duluth, MN-WI; Durham-Chapel Hill, NC; Edison-New Brunswick, NJ; Elmira, NY; Eugene-Springfield, OR; Fargo, ND-MN; Florence, SC; Fort Smith, AR-OK; Glens Falls, NY; Grand Forks, ND-MN; Grand Junction, CO; Great Falls, MT; Greenville-Mauldin-Easley, SC; Harrisburg-Carlisle, PA; Harrisonburg, VA; Hartford-West Hartford-East Hartford, CT; Honolulu, HI; Hot Springs, AR; Houma-Bayou Cane-Thibodaux, LA; Huntsville, AL; Iowa City, IA; Johnson City, TN; Kingsport-Bristol-Bristol, TN-VA; Lafayette, LA; Lake Charles, LA; Lancaster, PA; Lawrence, KS; Lebanon, PA; Longview, WA; Lynchburg, VA; Midland, TX; Missoula, MT; Mobile, AL; Mount Vernon-Anacortes, WA; Nassau-Suffolk, NY; New Orleans-Metairie-Kenner, LA; New York-White Plains-Wayne, NY-NJ; Ocean City, NJ; Philadelphia, PA; Pittsburgh, PA; Pittsfield, MA; Portland-Vancouver-Beaverton, OR-WA; Raleigh-Cary, NC; Reading, PA; Roanoke, VA; Salem, OR; San Angelo, TX; San Francisco-San Mateo-Redwood City, CA; San Jose-Sunnyvale-Santa Clara, CA; Seattle-Bellevue-Everett, WA; Sioux City, IA-NE-SD; Sioux Falls, SD; Spokane, WA; State College, PA; Trenton-Ewing, NJ; Tulsa, OK; Victoria, TX; Wenatchee-East Wenatchee, WA; Williamsport, PA; Wilmington, NC; Yakima, WA; York-Hanover, PA.

A.2 ADDITIONAL DATA DESCRIPTION

A.2.1 CRISM

We start with a 50% sample of all McDash (also known as LPS) mortgages linked to Equifax credit records that were outstanding for at least one month between January 2007 and December 2010. The CRISM data set provides the linked Equifax credit records for each McDash mortgage for the lifetime of the loan, including an additional 6 months before origination and after termination. This link is done directly by Equifax. Credit records provide a consumer's total outstanding debt amounts in different categories (first-lien mortgages, second-lien mortgages, home equity lines of credit [HELOCs], auto loans, etc.). Additionally, in any month, Equifax provides the origination date, amount, and remaining principal balance of the two largest (in balance terms) first mortgages, closed-end seconds, and HELOCs outstanding for a given consumer.

We convert these records into a loan-level panel with each loan's type, origination month, origination amount, termination month, and remaining principal balance for all months that the loan is outstanding. We restrict our sample to those consumers who start our sample with two or fewer loans in each category and never have more than three of any of these types of loans outstanding.¹ This amounts to about 96% of the population of Equifax borrower IDs, and these IDs cover about 90% of the loans in McDash. In creating this loan-level data set, we assume that the month in which the loan stops appearing in Equifax is the month that it was terminated.

The variables that McDash provides are already in the form of a loan-level panel and include: origination date, origination amount, remaining principal balance, termination date, termination type, lien type, interest type, property zip code, and purpose type. We match these to our Equifax panel. We consider an Equifax loan/McDash loan pairing a match if the origination date of the Equifax loan is within 1 month and the origination amount is within \$10,000 of the McDash loan. If more than one loan is matched, we use the origination amount, date, termination date, zip code, and termination balance as tiebreakers. We are able to match more than 93% of McDash loans using these restrictions, with more than 80% matching the origination information perfectly (up to \$1 in balances due to rounding).

We use the set of Equifax/McDash matched loans as our universe in our analysis. Owing to the restrictions above, this amounts to about 82% of the McDash universe. We also verify that we are correctly measuring the termination date and termination balance using the Equifax records by checking these variables against their McDash counterparts for the matched loans.

Over our period of interest (2008-9), our sample contains about 15 million mortgages per month; per MSA-month, we have on average 39,200 mortgages. One way to assess the coverage of our sample is to compare the outstanding mortgage balances by MSA to those in the FRBNY Consumer Credit Panel (CCP). The balances in our sample correspond on average to 29% of those in the CCP (with little variation over time), with a cross-sectional standard deviation across MSAs of 5% (the 25th percentile is at 26%, the 75th percentile at 32%). Since we start with a 50% random sample from CRISM and apply some filters as explained above, this means that CRISM covers over 60% of outstanding mortgages over this period.

A.2.2 *Measuring Refinancing Propensities*

Our goal is to measure the proportion of outstanding loans in an MSA that were refinanced in a given month. For the denominator, we start with all outstanding first liens (where lien type is measured using the McDash variable) in our Equifax/McDash matched universe, but exclude in each month loans that terminate in the next month because they were transferred to another servicer or terminate

¹This restriction allows us to infer the origination month, origination balance, and balance of the third largest loan of any loan type even though this information does not appear explicitly in Equifax, where if the third largest loan is also the newest loan, we assume its origination month to be the first month it appears in Equifax. We also drop loans that do not have complete consecutive Equifax records.

for unknown reasons (since we will be looking at the proportion of loans that are voluntarily paid off and refinanced).

We count a loan as being refinanced if: (1) its McDash termination type is a "voluntary payoff," and (2) for that consumer, there is another loan that is opened around the time of the first loan's termination on the same property (i.e. the new loan is a refinance, rather than a new purchase loan). More specifically, the most clear indicator that the new loan was a refinance is if the loan has a matching McDash loan (about 70%), and that McDash loan is marked as a refinance loan (in McDash's purpose type variable). On the other hand, the loan is clearly a new purchase loan if the purpose type is marked as such. However, about 25% of McDash loans have purpose type "Unknown" or "Other," and about 30% of the new loans are not matched in McDash (they only appear in Equifax, since McDash does not cover the entire market) and thus have no purpose type attached.

We thus use the following rules to identify refinances. We start by looking for any loan in the Equifax data set that has an open date within 4 months of the McDash loan's termination date. We find at least one such loan for about 81% of the voluntary terminations in 2008 and 2009. We classify these new loans as a refinance if either:

- The loan also appears in McDash and is tagged as a refinance in the purpose-type variable (61% of the McDash-matched loans).
- The loan also appears in McDash and is tagged as an "Unknown" or "Other" purpose type, and has the same property zip code as the original loan.
- The loan appears only in Equifax but the borrower's Equifax address does not change in the 6 months following the termination of the original loan.

This allows us to compute our measure of interest, the balance-weighted refinance propensity, as (balance outstanding in t-1 of loans that were refinanced in month t) / (balance outstanding in McDash in month t-1 that does not terminate for unknown reason in month t).

As a check, we calculate the refinance propensities separately for the three different cases above (McDash, known purpose; McDash, unknown purpose; Equifax), and find that these refinance propensities are very similar.

A.2.3 Measuring Cash-outs

To measure cash-out refinancings, we need to both identify refinances and how the balance of the new loan compares to the outstanding balance of the loan(s) paid off in the process.

We begin with Equifax/McDash first liens (again using the McDash lien type variable), and keep only those loans that have a McDash purpose type of refinance or unknown/other. Our algorithm to identify whether our new loan is a refinance is similar to the algorithm above. This time, we look for a loan (or loans) in Equifax that terminate(s) around the time when the new loan is originated and check that this loan looks like it was refinanced. We use McDash refinances rather than outstanding loans as our point of reference for these statistics so that we can better represent all refinances, rather than introducing potential bias through only seeing refinances of McDash loans.

Specifically, we call any loan in the Equifax data set that terminates between -1 and 4 months from our new loan's close date a "linked" loan, including first mortgages as well as closed-end seconds and HELOCs, and we call the new loan a refinance if:

- The loan is a known refinance in McDash. (For 86% of these, we find a linked loan in either McDash or Equifax. For the remaining 14%, we would consider these refinances where there was no previous loan on the property.)
- The loan has an "Unknown" or "Other" purpose type in McDash and a linked loan in McDash that has a matching property zip code.

- The loan has an "Unknown" or "Other" purpose type in McDash and a linked loan that appears only in Equifax, but the consumer's Equifax address does not change in the 6 months after the new loan was opened.

If there is more than one linked loan that is a first mortgage in Equifax, we link only the loan that is closest in balance to the origination amount of the new mortgage. We also allow to be linked only those Equifax loans that exist in the Equifax data for at least three months to prevent the refinanced loan balance from being counted in the old balance of the loan.

For each of these cases, we can then calculate the cash-out amount as the difference between the origination amount on the refinance loan and the balance of the linked loan(s) at termination. In order to capture the correct origination amount on the refinance loan, we want to ensure that we are also including any "piggyback" second liens that are opened with the refinance loan that we find in McDash. Thus, we look for any loan in the Equifax record linked to our refinance loan that has an Equifax open date within three months of our refinance loan and an origination balance of less than 25% of our loan's origination balance if labeled a first mortgage and less than 125% of the refinance loan's origination balance if labeled a HELOC or CES, and add the balance of these piggyback seconds to the refi origination amount when calculating cash-out amounts.² To eliminate outliers, we also drop cash-out and "cash-in" amounts that are greater than \$1,000,000. These amount to dropping less than 0.05% of the refinance loans.

At the MSA level, this allows us to calculate the amount cashed out relative to the total outstanding balance in month $t-1$. To estimate total dollar amounts cashed out, we scale up the amount cashed out by the ratio of total housing debt outstanding in an MSA according to the FRBNY Consumer Credit Panel (CCP) relative to the total outstanding balance in our CRISM sample. (The CCP amounts are available as end-of-quarter snapshots, so we interpolate between them to get a monthly series.)

In Figure A-9, we compare the total estimated quarterly cash-out amounts to those estimated on prime conventional loans by Freddie Mac.³ The figure shows that the two series co-move closely and also show similar levels. The higher level in CRISM is expected, since the Freddie Mac series does not include subprime/Alt-A as well as FHA and VA loans.

A.2.4 Measuring CLTVs/Equity

We start with all matched first-lien McDash loans. For a given month, we take the corresponding Equifax record and assign all outstanding second liens to the outstanding first liens in Equifax using the rule that each second lien is assigned to the largest first lien (in balance terms) that was opened on or before the second lien's opening date. We then add the assigned second lien balance(s) to the McDash balance of our original loan as our measure of secured debt on a property, which is the numerator of CLTV. For the denominator, the estimated updated property value, we start from the appraisal amount of the property at the time of the McDash loan origination and update it based on the local home price index from CoreLogic (using the zip-code-level index if available, and the MSA-level index if not).

Equity is simply defined as $1 - CLTV$. When taking medians within an MSA (our measure $E_{j,t}^{med}$), we weight individual observations by their current outstanding first-lien loan balance (from McDash).

A.2.5 Household-level Analysis of Car Spending Around Refinancing

In Section IV.D.1, we study the response of car purchases to refinancing at the borrower level. To do so, we start out with a random 50% sample of borrowers in our CRISM sample who refinance at some point over the year 2009. (We only use 50% for computational reasons.) We obtain monthly information

²We impose these upper bounds because we want to avoid picking up other first lien mortgages (to purchase another property) the borrower might originate at the same time.

³To make the two comparable, we multiply our CRISM total by $1/0.9175$, where 0.9175 is the share of mortgage balances in CRISM that is in MSAs (as opposed to micropolitan statistical areas or rural areas) as of November 2008.

on these borrowers' car loans from their Equifax credit records (we sum two categories, "auto bank"—loans provided by banks, credit unions, savings and loan associations—and "auto finance"—loans provided by automobile dealers and automobile financing companies) starting 6 months before the refinancing date until 12 months after. We only retain borrowers for whom there is only one refinancing in the data over this period. This leaves us with 676,425 refinancers.

We mark a borrower as obtaining a new car loan in a month if the total balance of car loans outstanding increases by \$2,000 or more (following Agarwal et al., 2017b); results are similar if we use different cutoffs. For Figure VI, we distinguish between non-cashout-refinances and cash-out refinances, where a refinance is categorized as the latter if 0.98 times the balance of the refinance loan (including simultaneously opened junior liens, as described in Section A.2.3) is at least \$5,000 above the balance of the old loan(s) that are paid off. Subtracting the 2 percent of the new loan amount is done in order to account for the fact that some of the cashed out amount may be used to cover costs associated with the refinancing. This threshold is close to the binary indicator used by GSEs, and conclusions are not sensitive to altering the cutoff. According to this metric, 44 percent of refinances in our 2009 sample were cash-outs.

A.2.6 HMDA

For robustness, in Figure A-5 we use a different measure of refinancing activity based on data made available as part of the Home Mortgage Disclosure Act (HMDA), which requires mortgage lenders to report information on mortgage applications and originations. The HMDA data are generally perceived to be the most comprehensive and representative source of information on mortgage applications and originations, with market coverage estimated to be around 90%.⁴ For each application, HMDA reports the geographic location of the property, the desired loan amount, the loan purpose (purchase or refinance), and whether the loan application led to an origination, was rejected by the lender, or was withdrawn by the borrower.⁵ While the public-use HMDA data contain only calendar year indicators, the private-use version of the data set (available to users within the Federal Reserve System) also contains the exact application date and the exact action date. The action date is the date on which the loan is originated, the application is rejected, or the application is withdrawn. These exact dates make the data suitable for high frequency event studies (see, for example, Fuster and Willen, 2010). In all our analysis using HMDA data, we retain only applications that led to originations (action code = 1), and always use the application date (rather than the action date). We drop multifamily properties and mortgages with an origination amount >\$3 million (about 0.015% of loans).

While the HMDA data are ideal for measuring the flow volume of mortgage origination activity across locations, it has two prominent limitations. First, for refinance loans, the HMDA data do not include any information on the loan that is paid off. As a result, we cannot use the HMDA data to estimate the extent to which households are removing cash from their mortgage during the refinancing process—a limitation we overcome with the CRISM data we focus on in the main paper. Second, the HMDA data do not include any information on the loans after they are originated. Thus, HMDA is not informative about how many outstanding mortgages there are in an MSA. The stock of outstanding mortgages is necessary to measure a refinancing propensity.

To obtain an estimate of the number of outstanding mortgages in each MSA, we supplement HMDA with data from the 2008 American Community Survey (ACS), which reports the number of outstanding mortgages (but not their amount) and the number of households for fine geographic areas. Since the ACS samples only a fraction of the population, we scale up the number of households based on Census information on the overall number of households in the US in 2008. We use the same scaling factor for the number of mortgages in each location. By combining the ACS data with the HMDA data, we can compute the number of loan originations per number of outstanding mortgages

⁴See, for instance Dell'Ariccia, Igan, and Laeven (2012).

⁵There are actually three designated loan types within the HMDA data: origination, refinancing, and home improvement. We combine the home improvement loans with the refinancing ones in our work below.

for each location within the U.S. For the analysis of refinance propensities around the 2001 recession, shown in Figure VIII in the main text, we use the same procedure but based on the 2000 Census.

The measures of MSA-level refinancing propensities in late 2008/early 2009 are highly correlated between the HMDA data and the CRISM data, once we account for the lag in CRISM relative to HMDA. The population-weighted cross-sectional correlation between the HMDA refinance propensity in December 2008 and the CRISM refinance propensity in January (February) 2009 is 0.87 (0.88). Pooling the second half of 2008 and the first half of 2009, the correlation between HMDA and CRISM propensities is 0.84 for the one-month-forward and 0.78 for the two-month-forward CRISM propensities.

A.2.7 Other Data

We now briefly describe some additional data used in our analysis. We measure unemployment rates for each MSA using data from the BLS's Local Area Unemployment Statistics. Our measure of labor income is based on the Individual Income Tax ZIP Code Data made available by the Internal Revenue Service. We first cumulate wage and salary incomes at the MSA level and then divide by the number of tax returns. We exclude tax returns from the top income category (>\$200,000) since those households can play a dominant role in driving average income changes; however, they are likely not the ones for which payment-to-income constraints (which we want to proxy for) are most relevant. Including those households, using adjusted gross income instead of wage and salary income, using the growth over 2007-2008 instead of 2008-2009, or using the income level instead of the growth rate does not change any of our conclusions in the regression tables below.

Our MSA-level demographic controls come from the ACS. We combine the 2007 and 2008 data to ensure the sample sizes are large enough to minimize measurement error. We measure each MSA's age composition, education composition, the percentage of homeowners, and the percentage of households with a mortgage. We restrict the ACS data to those individuals between the ages of 21 and 75 (inclusive) that were not living in group quarters (e.g., dorms, prisons, or medical facilities).

A.3 MODEL PROOFS AND DESCRIPTION OF NUMERICAL SOLUTION

We provide here the proof of the homogeneity of the value function as well as a description of our computational procedure.

To show the homogeneity property, we proceed by guess and verify. The value functions for refinancing and not refinancing are

$$\begin{aligned}
V^{noref_i}(\tilde{a}, \tilde{y}, \tilde{x}, r_0^m, r^m, r, F, 1) \tilde{p}^{\sigma-1} &= \max_{\{\hat{a}'\}} \frac{(\tilde{a}(1+r) + \tilde{y} - \gamma r_0^m \tilde{x} - \hat{a}')^{1-\sigma}}{1-\sigma} \tilde{p}^{\sigma-1} \\
&\quad + \beta \mathbb{E}[V(\hat{a}' \frac{\tilde{x}'}{\tilde{x}}, \tilde{y}', \tilde{x}', r_0^m, r^{m'}, r', F', 1) (\tilde{p}')^{\sigma-1} | \tilde{y}, \tilde{x}, r^m, r, F, 1] \\
V^{ref_i}(\tilde{a}, \tilde{y}, \tilde{x}, r_0^m, r, F, 1) \tilde{p}^{\sigma-1} &= \max_{\{\hat{a}'\}} \frac{(\tilde{a}(1+r) + \tilde{y} - \gamma r^m + \gamma(\tilde{x} - 1) - F - \hat{a}')^{1-\sigma}}{1-\sigma} \tilde{p}^{\sigma-1} \\
&\quad + \beta \mathbb{E}[V(\hat{a}' \frac{\tilde{x}'}{\tilde{x}}, \tilde{y}', \frac{\tilde{x}'}{\tilde{x}}, r^m, r^{m'}, r', F', 1) (\tilde{p}')^{\sigma-1} | \tilde{y}, \tilde{x}, r^m, r, F, 1]
\end{aligned}$$

Then we can eliminate $\tilde{p}$ as a state variable to obtain the transformed value function,

$$\begin{aligned}
J(\tilde{a}, \tilde{y}, \tilde{x}, r_0^m, r^m, r, F) &= \max\{J^{noref_i}(\tilde{a}, \tilde{y}, \tilde{x}, r_0^m, r^m, r, F), J^{ref_i}(\tilde{a}, \tilde{y}, \tilde{x}, r^m, r, F)\} \\
J^{noref_i}(\tilde{a}, \tilde{y}, \tilde{x}, r_0^m, r^m, r, F) &= \max_{\{\hat{a}'\}} \frac{(\tilde{a}(1+r) + \tilde{y} - \gamma r_0^m \tilde{x} - \hat{a}')^{1-\sigma}}{1-\sigma}
\end{aligned}$$

$$\begin{aligned}
& + \beta e^{(1-\sigma)(\mu_q - \pi)} \mathbb{E} \left[J(\hat{a}' \frac{\tilde{x}'}{\tilde{x}}, \tilde{y}', \tilde{x}', r_0^m, r^{m'}, r', F') | \tilde{y}, \tilde{x}, r^m, r, F \right] \\
J^{refi}(\tilde{a}, \tilde{y}, \tilde{x}, r^m, r, F) = & \max_{\{\hat{a}'\}} \frac{(\tilde{a}(1+r) + \tilde{y} - \gamma r^m + \gamma(\tilde{x} - 1) - F - \hat{a}')^{1-\sigma}}{1-\sigma} \\
& + \beta e^{(1-\sigma)(\mu_q - \pi)} \mathbb{E} \left[J(\hat{a}' \frac{\tilde{x}'}{\tilde{x}}, \tilde{y}', \frac{\tilde{x}'}{\tilde{x}}, r^m, r^{m'}, r', F') | \tilde{y}, \tilde{x}, r^m, r, F \right]
\end{aligned}$$

In order to solve the transformed value function, we discretize $\log \tilde{x}$ using 60 grid points evenly spaced with width μ_q between 0.45 and -1.025, 64 grid points for $\tilde{a}$ between 0 and 1, with more grid points near the lower asset values to account for the concavity of the value function, and 46 grid points for $\tilde{y}$ evenly spaced in logs between -0.5 and 0.5. The stochastic shock can take on 3 values: -1SD, 0, and +1SD with probabilities computed using the Tauchen algorithm. Households assume that interest rates will remain constant in the future, and the model is solved with both $r^m = 0.06$ and $r^m = 0.05$. The short-term interest rate r is permanently set to 0.03. The model is then solved using value function iteration. Finally, the model is simulated using 50,000 households and 9 regions. Households are initialized at median income and no assets in a middle income, middle house price region and 300 years of the model are discarded as an initial burn-in. Using a longer burn-in period or changing initial conditions prior to the burn-in period leaves results unchanged.

A number of alternative models are explored in Sections VII. and VIII. which involve changes to the baseline computational setup.

In the version of the model with ARMs, we assume that the households with ARMs' current payment moves one-for-one with the current mortgage rate when not refinancing. In particular, the J^{norefi} is identical except that r_0^m is always equal to r^m .

We calibrate the ARM share in bad regions to 30 percent, the ARM share in the middle regions to 20 percent and in the good regions to 10 percent. As described in the text, this is a conservative calibration for assessing the importance of ARMs since it overstates the actual variation in interest rate flexibility across space in 2008 as most ARMs do not have substantive rate resets during the QE episode.

We next explore a version of the model in which the housing bust in 2008 is preceded by a house price boom. Specifically, we assume that in the period before the Great Recession, there is an aggregate house price increase of 10% and a regional shock of $\pm 7.5\%$ that is perfectly negatively correlated with the shock during the bust. This roughly captures house price movements in the last year of the housing boom. We do not recalibrate parameters in this exercise to match the effects of QE1, but they are similar to the baseline.

We then move to a model which partially endogenizes the cash-out decision when refinancing. In particular, we let households choose between two different options when refinancing: 1) Extracting all of their housing equity when refinancing, as in our baseline. 2) Extracting no equity and only resetting the rate. In this model, we now have three value functions: J^{norefi} , J^{refi} and $J^{refi \text{ rate only}}$. $J^{refi \text{ and cash-out}}$ is identical to J^{refi} in the original problem (except with a continuation value that maximizes over three rather than two value functions). When doing a pure rate-refi, the mortgage balance is unchanged but the household pays a fixed cost to change their fixed rate to the current mortgage rate in the economy. This means that $J^{refi \text{ rate only}}$ is identical to J^{norefi} except that the rate adjusts to the current mortgage rate and the household must pay the refinancing cost. Note also that this value function is identical to that for the ARM model above, except that the household must pay the fixed cost to obtain the current rate instead of obtaining it for free:

$$J^{refi \text{ rate only}}(\tilde{a}, \tilde{y}, \tilde{x}, r^m, r, F) = \max_{\{\hat{a}'\}} \frac{(\tilde{a}(1+r) + \tilde{y} - \gamma r^m \tilde{x} - F - \hat{a}')^{1-\sigma}}{1-\sigma}$$

$$+ \beta e^{(1-\sigma)(\mu_q - \pi)} \mathbb{E} \left[J(\hat{a}' \frac{\tilde{x}'}{\tilde{x}}, \tilde{y}', \tilde{x}', r^m, r^{m'}, r', F') | \tilde{y}, \tilde{x}, r^m, r, F \right]$$

Also note that this model introduces no new parameters, so the calibration strategy is identical to that in the baseline model. Ultimately, few households in the model choose to do rate only-refinancing with no cash-out even when they have this option, because the general equity accumulation and extraction problem still plays a primary role in decisions. A household who is contemplating a rate only refinance today still knows that in the future, they will eventually want to extract equity. Since the fixed cost of refinancing is the same whether they extract equity or not today, it makes sense to extract equity when refinancing to secure the low rate rather than doing a rate refi today and then paying the fixed cost again in the near future to extract equity.

Since we write the above value functions for general interest rate processes, the statement of the problem with stochastic interest rates is identical to that in the baseline model with a one-time unanticipated shock. Computationally, in the baseline model with a one-time shock, we only need to solve the model for two values of r^m to compute impulse responses, and we do not need to calculate expectations over r^m . In the model with stochastic mortgage rates, r^m takes on 5 values with probabilities computed using the Tauchen algorithm, and this state-variable enters expectations in the value function.

A.4 APPENDIX TABLES AND FIGURES

This section provides additional tables and figures referred to in the main text.

- Table A-1 provides descriptive statistics on the 381 MSAs in our sample, including all the local control variables used in regressions reported in Tables I as well as Tables A-2 to A-5.
- Table A-2 documents the correlations between $E_{Nov2008}^{med}$ and other local controls.
- Table A-3 provides the results from regressions of refinancing propensities on $E_{Nov2008}^{med}$ and other local controls, all interacted with a post-QE indicator (equation (1)), but at the state level rather than the MSA level as in Table I.
- Table A-4 provides the results from regressions of cash-out shares on $E_{Nov2008}^{med}$ interacted with post-QE indicator and other local controls (including refinancing propensities in the last column), discussed in Section IV.D.
- Table A-5 provides the results from regressions of log(auto sales) on $E_{Nov2008}^{med}$ interacted with post-QE indicator, refinancing propensities and cash-out shares, discussed in Section IV.D (footnote 21).
- Figure A-1 shows that refinancing activity in HMDA data follows similar patterns to those shown in Figure I, and that there was no discernible increase in purchase mortgage activity around QE1.
- Figure A-2 shows that $E_{Nov2008}^{med}$ is highly correlated with local shares of borrowers that have $CLTV > 80$ or $CLTV > 100$, illustrating that the equity of the median borrower correlates strongly with other moments of the equity distribution, as discussed in Section IV.B.
- Figure A-3 shows distributions of MSA-median and individual-level equity in January 2007 and November 2008, as discussed in Section IV.B.
- Figure A-4 shows the relationships between $E_{Nov2008}^{med}$ and local house price growth over Jan 2007-Nov 2008, and between $E_{Nov2008}^{med}$ and changes in the local unemployment rate over Jan 2007-Nov 2008, as discussed in Section IV.B.
- Figure A-5 shows that the patterns shown in Figure III are very similar when using the HMDA data, where we can sum originated refinance mortgages by their application date. The figures show that activity jumped right after the QE1 announcement, and more so in high equity MSAs.
- Figure A-6 shows that there was essentially no variation in rates on new mortgages between high and low equity MSAs, as discussed in Section IV.B.
- Figure A-7 shows that there was no bunching in PTI ratios and no differences across high and low equity MSAs in the PTI distribution in 2009, while there is strong bunching and differences across MSA groups in CLTVs, as discussed in Section IV.B.
- Figure A-8 shows that non-housing debt increased in high equity MSAs relative to low equity MSAs following QE1, as discussed in Section IV.D.
- Figure A-9 shows that cash-out amounts in CRISM comove closely with those in Freddie Mac data, as discussed in Appendix A.2.3.
- Figure A-10 shows that the household-level distribution of equity in the 2008 calibration of our model is a good fit to the empirical distribution shown in Figure A-3, as discussed in Section VI.C.
- Figure A-11 shows how the refinancing threshold in our model changes with assets and earnings.

- Figure A-12 shows the separate contributions of borrowers and lenders to the aggregate consumption impulse response in the 2008 baseline model shown in Figure XI.
- Figure A-13 shows that shifts in the mean of the equity distribution have effects on monetary policy that depend on the variance of the distribution, as discussed in Section VI.E.2.
- Figure A-14 shows the consumption response in the model under different assumptions about the share of lenders that are in the domestic economy, as discussed in Section VIII.A.
- Figure A-15 shows the consumption response in a version of the model where income is partly endogenous to consumption, as discussed in Section VIII.A.
- Figure A-16 shows how ARM shares as of November 2008 and ARM payment reductions over Nov 2008-June 2009 are related to $E_{Nov2008}^{med}$, as discussed in Section VIII.B.1.
- Figures A-17-A-19 show results from additional robustness checks discussed in Section VIII..

Table A-1: Descriptive Statistics. Data on 381 MSAs; statistics are unweighted.

	Mean	SD	p10	p25	p50	p75	p90
Median equity share (E^{med}), Jan 2007	0.293	0.066	0.209	0.243	0.289	0.345	0.383
Median equity share (E^{med}), Nov 2008	0.160	0.110	0.015	0.120	0.190	0.230	0.261
Δ House Price Index, Jan 2007-Nov 2008	-0.111	0.113	-0.286	-0.161	-0.086	-0.033	0.010
Δ Unemployment, Jan 2007-Nov 2008	2.297	1.370	0.700	1.400	2.100	3.000	4.100
Δ Labor Income, 2008-2009	0.003	0.022	-0.022	-0.009	0.003	0.017	0.025
<i>Average mortgage characteristics as of Nov 2008 (based on CRISM):</i>							
FICO score	701.8	17.4	679.4	689.4	701.6	714.9	723.5
Current interest rate (%)	6.23	0.16	6.02	6.13	6.23	6.34	6.43
Loan age (months)	39.3	3.8	34.6	36.6	39.1	41.6	44.8
Share jumbos (based on current balance)	0.046	0.048	0.007	0.014	0.029	0.063	0.115
Average balance of non-jumbo loans (1000s)	130.7	46.2	87.3	100.0	117.3	153.2	181.2
Share adjustable-rate mortgages	0.136	0.098	0.057	0.070	0.098	0.171	0.272
Share GSE securitized	0.615	0.097	0.498	0.572	0.631	0.676	0.721
Share privately securitized	0.158	0.086	0.083	0.103	0.127	0.180	0.262
<i>Local demographics (based on ACS):</i>							
Years of education (shares)							
12	0.147	0.033	0.105	0.124	0.146	0.170	0.192
13-15	0.315	0.042	0.262	0.287	0.316	0.339	0.371
16	0.172	0.048	0.112	0.135	0.168	0.203	0.234
>16	0.266	0.026	0.236	0.250	0.263	0.281	0.300
Age group (shares)							
31-45	0.286	0.026	0.252	0.271	0.285	0.303	0.318
46-40	0.340	0.024	0.311	0.326	0.340	0.356	0.370
>60	0.206	0.038	0.167	0.183	0.202	0.223	0.244
Homeowners (share)	0.779	0.053	0.712	0.751	0.785	0.813	0.839
Mortgage borrowers (share)	0.543	0.065	0.454	0.505	0.552	0.591	0.619

Table shows descriptive statistics for all 381 MSAs in our analysis sample, including the distribution of median equity shares (E^{med}) in January 2007 and November 2008, the distribution of house price and unemployment rate changes (in percent and percentage points, respectively) between January 2007 and November 2008, the distribution of changes in average labor income between 2008 and 2009, distributions of a number of characteristics of the outstanding mortgages in each MSA as of November 2008 (taking balance-weighted averages within each MSA), and local demographic characteristics as of 2008. For years of education and age groups, the base categories (<12 years of education; ages 21-30) are not shown. The share of mortgage borrowers is relative to all households, not just homeowners.

Table A-2: Relationships between Median Equity and Unemployment Increases as well as Various Average Characteristics of Outstanding Mortgages

	(1)	(2)	(3)	(4)	(5)	(6)
Δ Unemployment, Jan 2007-Nov 2008	-0.0597*** (0.00877)					-0.00236 (0.00434)
Δ Labor Income, 2008-09		3.705*** (0.513)				0.423 (0.257)
FICO score			0.00404*** (0.000661)			0.00528*** (0.000627)
Current interest rate (%)				-0.362*** (0.110)		0.347*** (0.0662)
Loan age (months)				0.00238 (0.00413)		-0.000393 (0.00142)
Share jumbos (based on current balance)				-0.473** (0.227)		0.661*** (0.125)
Average balance (non-jumbo)				-0.0189 (0.0467)		0.169*** (0.0280)
Share adjustable-rate mortgages					-1.128*** (0.321)	-0.942*** (0.150)
Share GSE securitized					0.0793 (0.124)	-0.437*** (0.0643)
Share privately securitized					0.956** (0.409)	-0.642*** (0.217)
Constant	0.302*** (0.0208)	0.159*** (0.0107)	-2.690*** (0.467)	2.365*** (0.766)	0.126 (0.0910)	-4.621*** (0.761)
Demographics	N	N	N	N	N	Y
N	381	381	381	381	381	381
Adj. R^2	0.36	0.31	0.30	0.16	0.17	0.85

Table shows the results of regressions of median equity as of November 2008 ($E_{i,Nov2008}^{med}$) on various other variables observed as of November 2008 (and shown in Table A-1). Sample consists of 381 MSAs. MSAs are weighted by their 2008 population. Robust standard errors in parentheses. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A-3: Responses of Local Refinancing to Equity: State-Level Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$E_{Nov2008}^{ind} \times \text{postQE}$	1.788*** (0.551)	1.379** (0.573)	1.577** (0.726)	1.214** (0.525)	0.887*** (0.126)	2.099*** (0.483)	1.528*** (0.297)	0.895** (0.371)
$\Delta UR_{Jan07-Nov08} \times \text{postQE}$		-0.0602 (0.0451)						-0.0145 (0.0257)
$\Delta Income_{2008-09} \times \text{postQE}$			3.283 (4.314)					-2.675 (2.745)
FICO $\times$ postQE				0.0114*** (0.00241)				0.00371 (0.00390)
Current int. rate $\times$ postQE					-2.006*** (0.185)			-1.453*** (0.478)
Loan age $\times$ postQE					-0.0348*** (0.00705)			-0.0301** (0.0113)
Jumbo share $\times$ postQE					-2.630*** (0.595)			-2.656** (1.237)
Average balance (non-jumbo) $\times$ postQE					-0.294*** (0.0434)			-0.211 (0.164)
ARM share $\times$ postQE						3.051*** (0.870)		0.295 (0.833)
GSE share $\times$ postQE						1.244* (0.684)		0.504 (0.690)
Private sec. share $\times$ postQE						-2.640* (1.327)		-0.0753 (1.282)
Demographics $\times$ postQE	N	N	N	N	N	N	Y	Y
State fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Month fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Adj. R^2	0.90	0.90	0.90	0.91	0.93	0.91	0.92	0.93
Adj. R^2 (within)	0.19	0.21	0.19	0.29	0.41	0.29	0.32	0.41
Observations	600	600	600	600	600	600	600	600

Table repeats the results from Table 1 but using state rather than MSA level refinancing and median equity. States are weighted by their 2008 population. Robust standard errors (clustered by state) in parentheses. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A-4: The Responsiveness of Cash-Out Refinancing Activity to Equity and Other Local Characteristics around QE1

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$E_{Nov2008}^{fixed} \times \text{postQE}$	0.167*** (0.0224)	0.138*** (0.0266)	0.147*** (0.0271)	0.103*** (0.0233)	0.123*** (0.0177)	0.171*** (0.0248)	0.157*** (0.0189)	0.0458** (0.0207)	0.0626*** (0.0137)
$\Delta UR_{Jan07-Nov08} \times \text{postQE}$		-0.00479*** (0.00176)						-0.00414*** (0.00145)	
$\Delta Income_{2008-09} \times \text{postQE}$			0.236** (0.117)					0.0303 (0.0906)	
FICO $\times \text{postQE}$				0.000862*** (0.0000968)				0.000297 (0.000227)	
Current int. rate $\times \text{postQE}$					-0.115*** (0.0146)			-0.112*** (0.0231)	
Loan age $\times \text{postQE}$					-0.00140*** (0.000434)			-0.00219*** (0.000567)	
Jumbo share $\times \text{postQE}$					-0.147*** (0.0398)			-0.0226 (0.0472)	
Average balance (non-jumbo) $\times \text{postQE}$					-0.00633 (0.00537)			-0.00454 (0.0123)	
ARM share $\times \text{postQE}$						0.00399 (0.0493)		-0.230*** (0.0435)	
GSE share $\times \text{postQE}$						0.131*** (0.0291)		0.0437* (0.0251)	
Private sec. share $\times \text{postQE}$						0.0990 (0.0663)		0.266*** (0.0667)	
Refinancing propensity									0.0610*** (0.00331)
Demographics $\times \text{postQE}$	N	N	N	N	N	N	Y	Y	N
MSA fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Month fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Adj. R^2	0.80	0.80	0.80	0.81	0.81	0.80	0.80	0.82	0.84
Adj. R^2 (within)	0.12	0.12	0.12	0.15	0.17	0.13	0.14	0.20	0.32
Observations	4572	4572	4572	4572	4572	4572	4572	4572	4572

Table shows results from regressions of monthly equity removed (relative to outstanding balances) in 381 MSAs from August 2008 to July 2009, measured in CRISM, on median equity in each MSA interacted with a dummy "postQE" that equals one for the period February–July 2009, and other variables also interacted with postQE. Demographic controls in columns 7 and 8 are location-specific shares of individuals with different years of education: 12,13-15,16,>16 (<12 is omitted), shares of individuals with ages: 31-45,46-60,>61 (21-30 is omitted), homeowner shares and the share of households with mortgages, from the ACS. Each of these demographic controls is interacted with "postQE" to allow for differential responses to rate declines. All regressions also include MSA and month fixed effects. For reference, the average monthly equity removed over August 2008–January 2009 was 0.09 percent of outstanding balances, while over February–July 2009 it was 0.14 percent of outstanding balances. Descriptive statistics of explanatory variables are provided in Table A-1. MSAs are weighted by their 2008 population. Robust standard errors (clustered by MSA) in parentheses. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

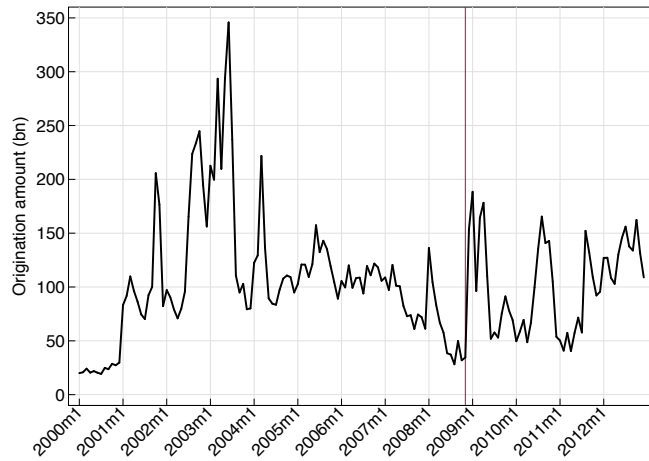
Table A-5: The Responsiveness of New Vehicle Spending around QE1

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$E_{Nov2008}^{med} \times \text{postQE}$	0.389*** (0.0851)	0.246** (0.113)	0.227*** (0.0574)					0.104 (0.129)	-0.174 (0.144)
$E_{Nov2008}^{med} \times \text{time trend (monthly)}$		0.0103 (0.00918)	0.0110*** (0.00316)					0.0120 (0.00982)	0.0118 (0.00969)
(Refinancing propensity) _{t-2}				0.0994*** (0.0244)	0.116*** (0.0206)			0.0874*** (0.0212)	0.0809*** (0.0227)
(Cashout/balance) _{t-2}						0.485*** (0.165)	0.369*** (0.0923)	0.0511 (0.0621)	0.00689 (0.0513)
Other cont. $\times$ postQE	N	N	N	N	N	N	N	N	N
MSA fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Month fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Adj. R ²	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Adj. R ² (within)	0.04	0.03	0.05	0.04	0.04	0.01	0.01	0.05	0.09
Observations	4572	9144	18288	4572	9144	4572	9144	9144	9144
Date range	200808.200907	200801.200912	200701.201012	200808.200907	200801.200912	200808.200907	200801.200912	200801.200912	200801.200912

Column 1 shows results from regression of monthly log(auto sales) in 381 MSAs on median equity ($E_{Nov2008}^{med}$) in each MSA (as of November 2008) interacted with a dummy "postQE" that equals one from February 2009 onward. In columns 2 and 3, the sample period is extended and a monthly trend interacted with $E_{Nov2008}^{med}$ is added. In columns 4 and 5, log(auto sales) is directly regressed on two-months-lagged local refinancing propensities; in columns 6 and 7 the same is repeated for local cash-out amounts (relative to outstanding balance). Columns 8 and 9 add $E_{Nov2008}^{med} \times \text{postQE}$ and both refinancing measures simultaneously; in column 9 we further add all mortgage market and demographic controls interacted with postQE (as in the last column of Table 1). All regressions also include MSA and month fixed effects. MSAs are weighted by their 2008 population. Robust standard errors (clustered by MSA) in parentheses. Auto sales are from R.L. Polk; refinancing propensities and cash-out fractions are measured in CRISM. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure A-1: Mortgage Activity in the US over 2000-2012, Based on HMDA Data

(a): Refinance Mortgage Originations



(b): Purchase Mortgage Originations

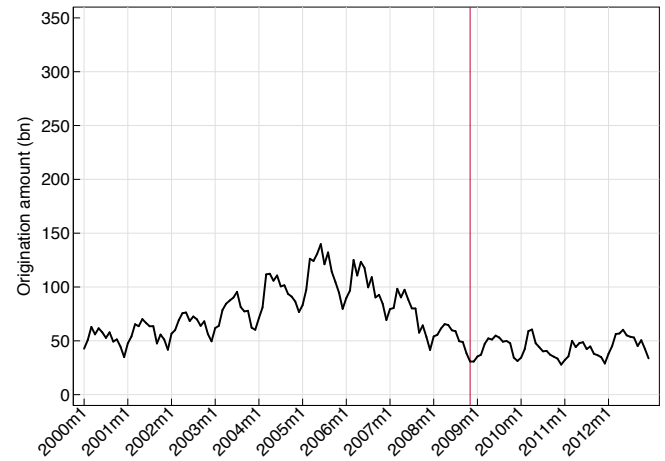


Figure shows mortgage originations on 1-4 unit homes in HMDA, by month in which the borrower applied for the loan. The vertical line indicates the month of the QE1 announcement (November 2008).

Figure A-2: Median Equity vs. Fraction of Borrowers over CLTV 80 or CLTV 100 Thresholds, as of November 2008.

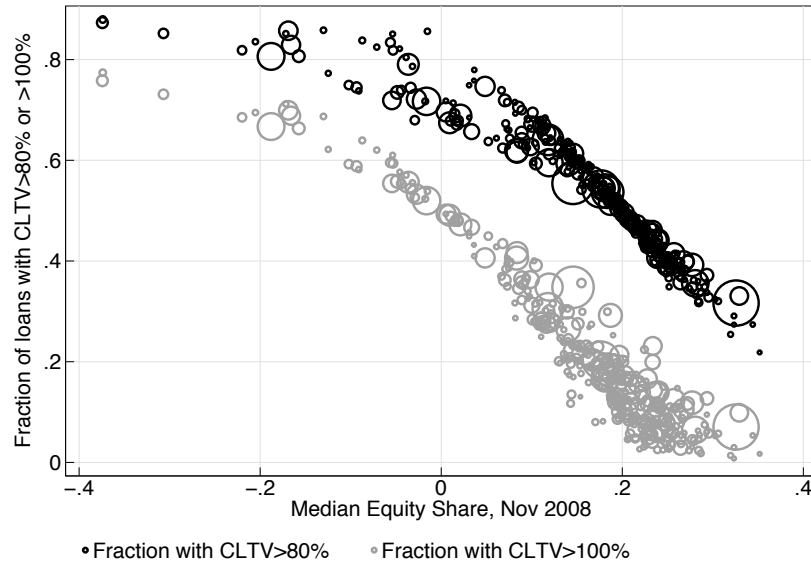
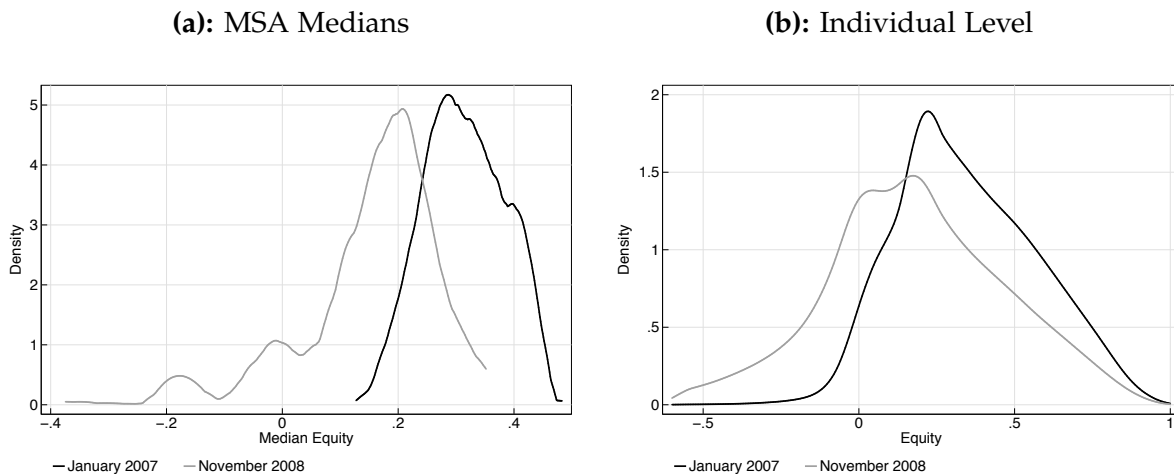


Figure shows the relationship between median equity in an MSA as of November 2008 ($E_{j,Nov2008}^{med}$) and the share of borrowers in the MSA who have an estimated CLTV in the same month higher than 80 percent (black circles) or higher than 100 percent (gray circles). Equity/CLTVs are measured based on CRISM. Each observation is an MSA, with 381 in total (per series). The size of the circle represents the 2008 population of the MSA.

Figure A-3: Distributions of Borrowers' Equity in their Homes — MSA Medians and Individual Level

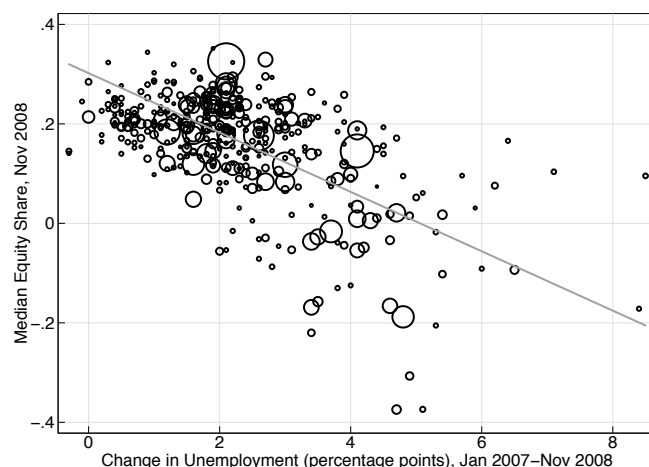
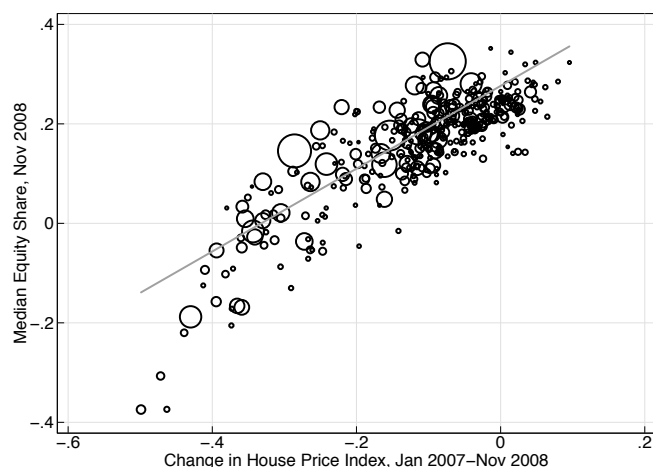


Panel (a) shows kernel density of $E_{j,t}^{med}$ across 381 MSAs in January 2007 and November 2008; MSAs are weighted by their 2008 population. Panel (b) shows kernel density of individual borrower equity in January 2007 and November 2008; borrowers are weighted by loan amount. Individual equity is more disperse than median equity at the MSA level because individual variation has life-cycle and other idiosyncratic components that are large relative to cross-region variation.

Figure A-4: Relationship between Equity and Other Measures of Economic Activity.

(a): House Price Growth vs. Median Equity

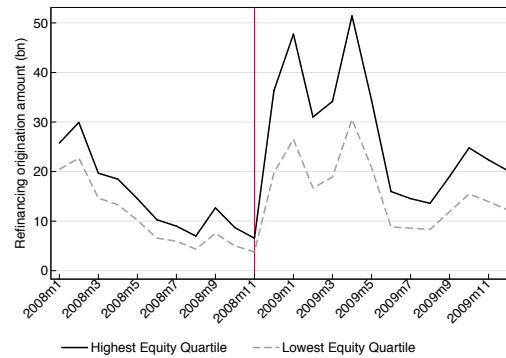
(b): Unemployment Rate Change vs. Median Equity



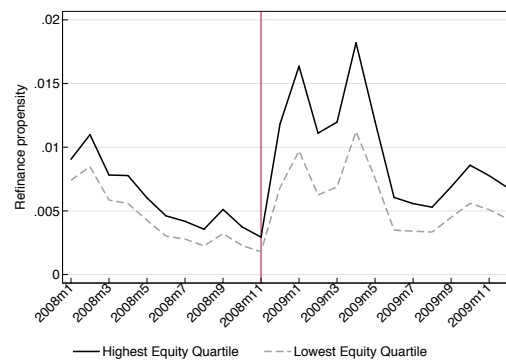
Panel (a) shows MSA house price growth between January 2007 and November 2008 vs. the median borrower's equity (as a share of estimated home value) in the MSA in November 2008. Each observation is an MSA, with 381 in total. The size of the circle represents the 2008 population of the MSA. The figure also shows the simple (population weighted) regression through the scatter plot: a 1 percent decline in house prices is associated with a 0.83 percentage point decrease in median equity (standard error 0.07) with an R-squared of 0.69. Panel (b) shows the change in an MSA's unemployment rate between January 2007 and November 2008 vs. the median borrower's equity (as a share of estimated home value) in the MSA in November 2008. The simple regression line shows that a 1 percentage point increase in the unemployment rate is associated with a 6.0 percentage point decline in median equity (standard error 0.09) with an R-squared of 0.36.

Figure A-5: Mortgage Refinance Activity 2008-2009 in Top and Bottom Quartile of MSAs Defined by Median Equity in November 2008, Based on HMDA Data

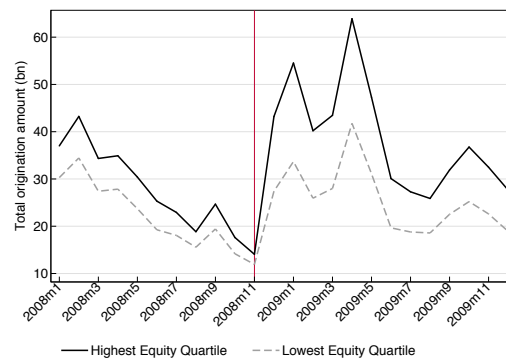
(a): Refinance Origination Volumes



(b): Refinance Propensities



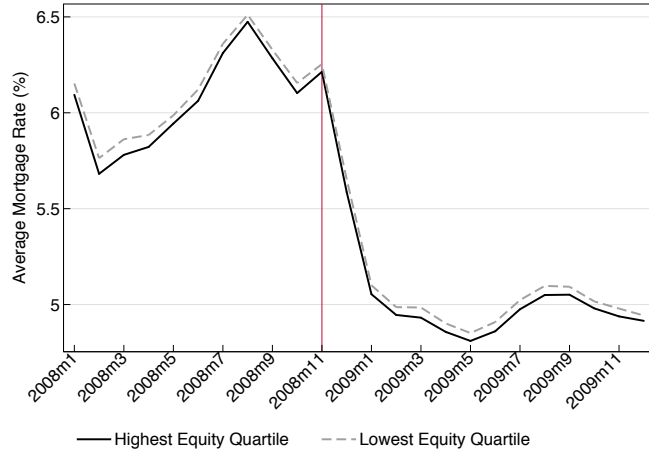
(c): Total Origination Volumes (Refinance and Purchase)



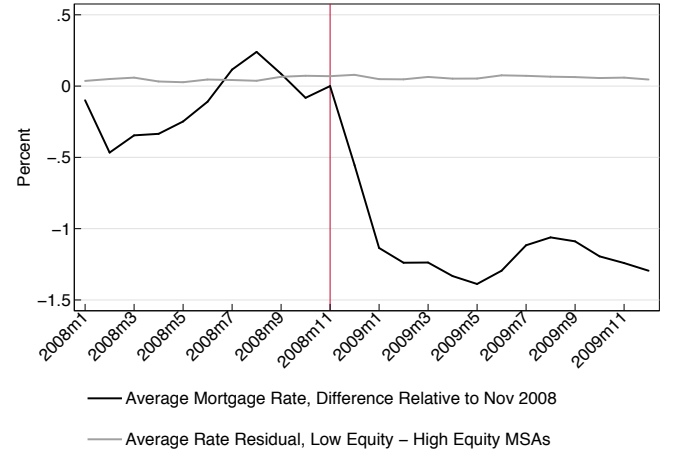
Panel (a) shows total mortgage refinance volume in HMDA by month in which borrower applied for the mortgage, where months are re-defined such that they start on the 25th day of the prior month. Panel (b) shows corresponding refinance propensities, defined as the number of refinance originations in HMDA divided by the total number of mortgages outstanding as measured in the 2008 American Community Survey. Panel (c) shows total mortgage origination volume (purchase and refinance) in HMDA by month in which borrower applied for the mortgage. In all three panels, calculations are done at the level of MSA quartile groups and vertical lines indicate the month of the QE1 announcement (November 2008).

Figure A-6: No Spatial Variation in Mortgage Interest Rates

(a): Raw Interest Rates by Equity Quartile



(b): Interest Rate Gaps After Controlling for Observables



Panel (a) shows average interest rate on newly originated non-purchase mortgages in CRISM in the top and bottom MSA quartiles by $E_{j,Nov2008}^{med}$, showing that the evolution of mortgage rates was almost identical across locations. In panel (b), the black line shows the difference in the average interest rate on newly originated mortgages relative to its level in November 2008. The gray line shows the gap between average rate residuals in the top and bottom MSA quartiles by $E_{j,Nov2008}^{med}$. Rate residuals are obtained following [Hurst et al. \(2016\)](#), by regressing each newly originated mortgage's interest rate on quadratic functions of FICO and LTV, and dummies for different loan purposes (purchase, rate refi, cashout refi, unknown), all interacted with month dummies. The gap averages only around 5 basis points over our sample and does not increase after QE1.

Figure A-7: 2009 Distributions of PTI and CLTV for Top and Bottom Quartile of MSAs Defined by Median Borrower Equity in November 2008

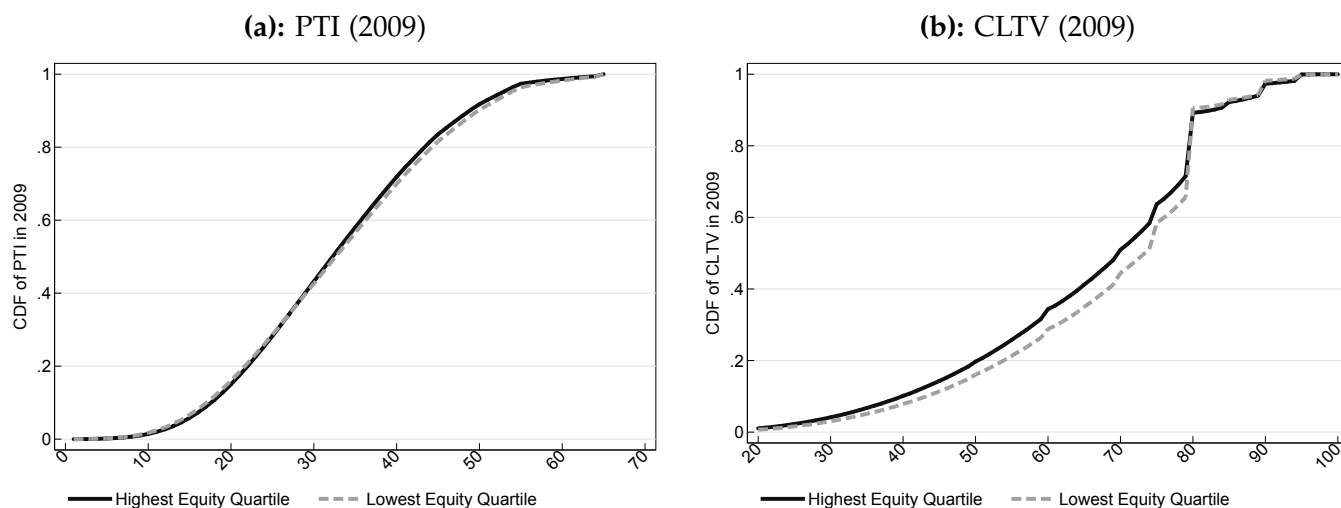


Figure shows the distribution of PTI and CLTV at origination in high and low equity MSA groups for loans originated in 2009 using the pooled sample of loans acquired by Fannie Mae and Freddie Mac in Single Family Loan Performance (SFLP) Data. In total, we observe 1,886,365 originations in the highest equity quartile and 1,514,085 originations in the lowest equity quartile. We set the 48,537 loans with missing PTI equal to 0.65 since SFLP does not report PTI above 0.65, but results are similar when instead excluding these loans. Panel (a) shows that the distribution of PTI exhibits little bunching and is similar in both high and low equity MSAs (as listed in Appendix A.1). Panel (b) shows that the distribution of CLTV (for the flow of new mortgages in SFLP) exhibits substantial bunching and that CLTVs on newly originated loans tend to be higher in MSAs where borrowers have less equity in outstanding mortgages (as measured in CRISM).

Figure A-8: Evolution of Non-Housing Debt for Top and Bottom Quartile of MSAs Defined by Median Borrower Equity in November 2008

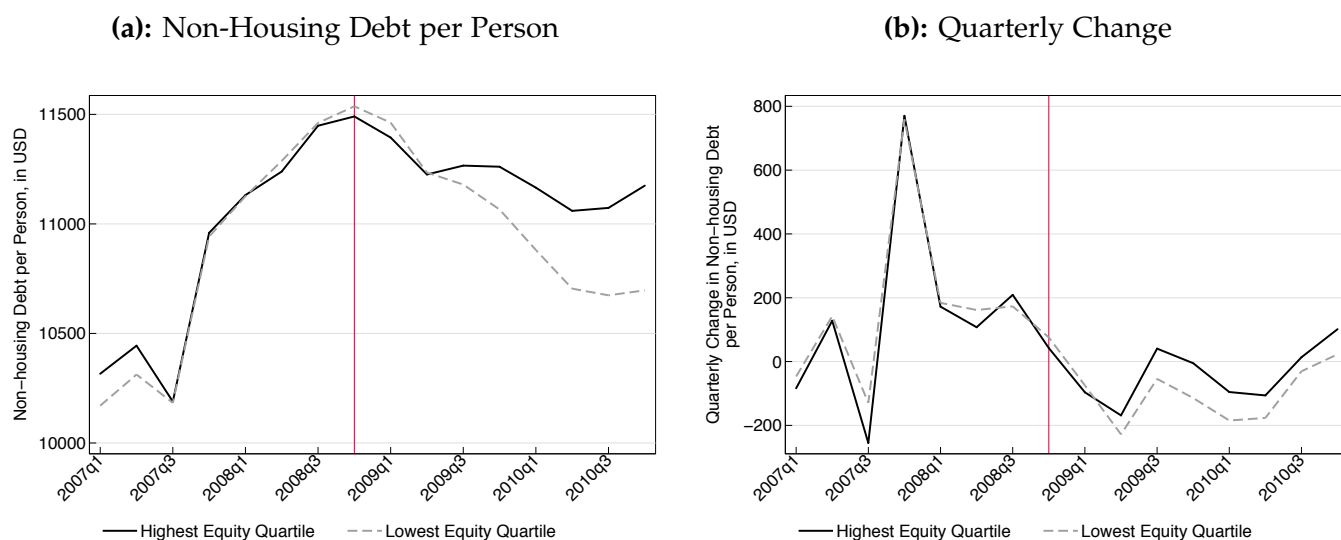


Figure shows level (panel (a)) and quarterly change (panel (b)) of non-housing debt per person in the FRBNY Consumer Credit Panel in the top and bottom MSA quartiles by $E_{j,Nov2008}^{med}$. Non-housing debt includes auto loans, credit card balances, student loans, consumer finance loans (sales financing, personal loans), retail loans (department stores, home furnishings, gas, etc.), and unspecified "other" loans.

Figure A-9: Estimated Cash-out Amounts from Freddie Mac vs. in our CRISM data

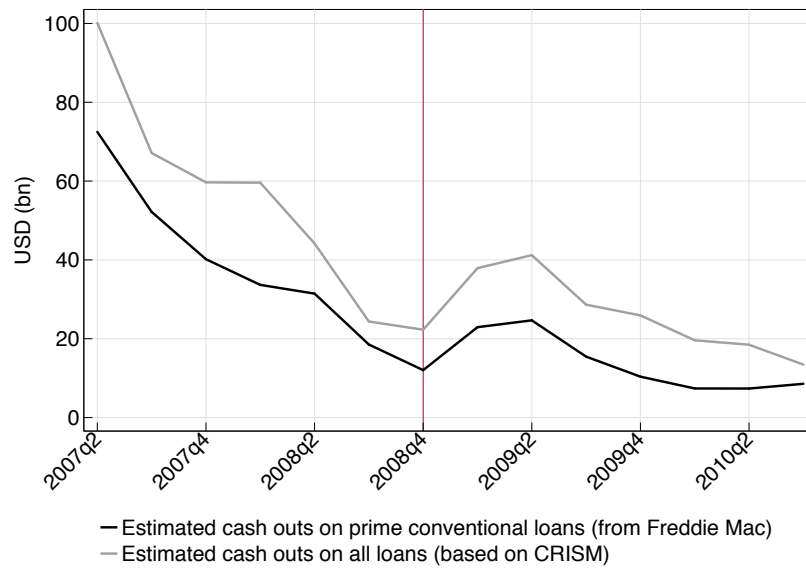
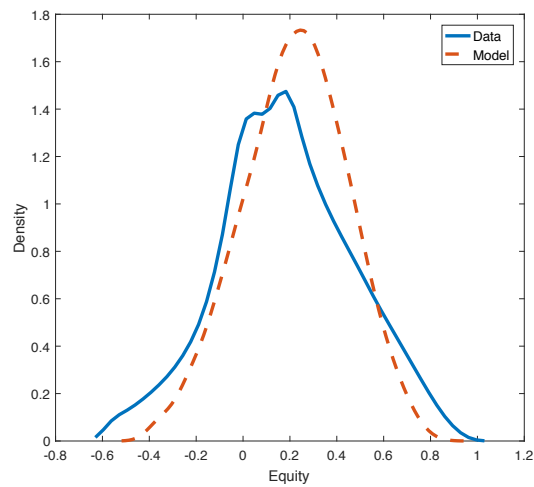


Figure shows estimated quarterly cash-out volumes on prime conventional (non-government) mortgages estimated by Freddie Mac (obtained from http://www.freddiemac.com/finance/docs/q4_refinance_2014.xls), as well as those we obtain based on the CRISM data (which also include FHA/VA loans) after scaling up as explained in Section A.2.3.

Figure A-10: 2008 Household Equity Distribution: Model vs. Data



This figure compares the household level density of equity in 2008 with no reduction in interest rates to the empirical distribution just prior to QE1 in November 2008.

Figure A-11: Relationship between Refinancing Threshold, Equity and Income

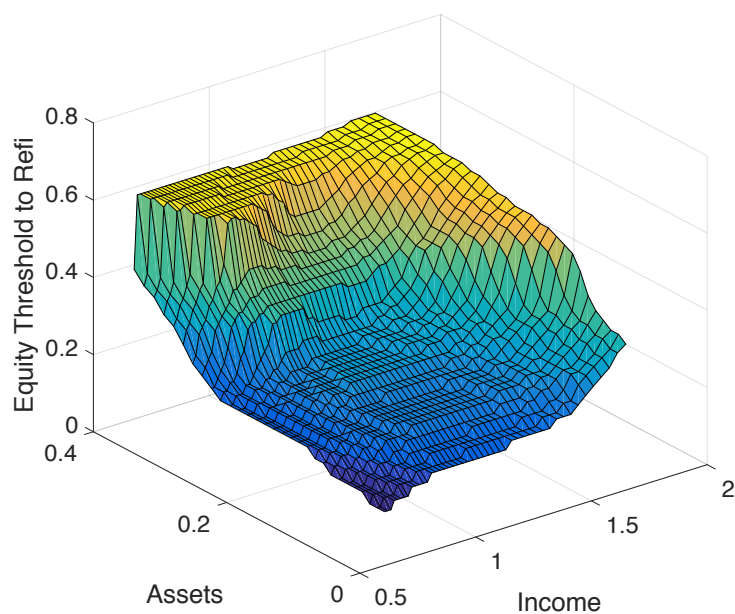


Figure shows refinancing threshold for each asset and income value for low fixed cost.

Figure A-12: Borrower and Lender Contribution to Aggregate Consumption Impulse

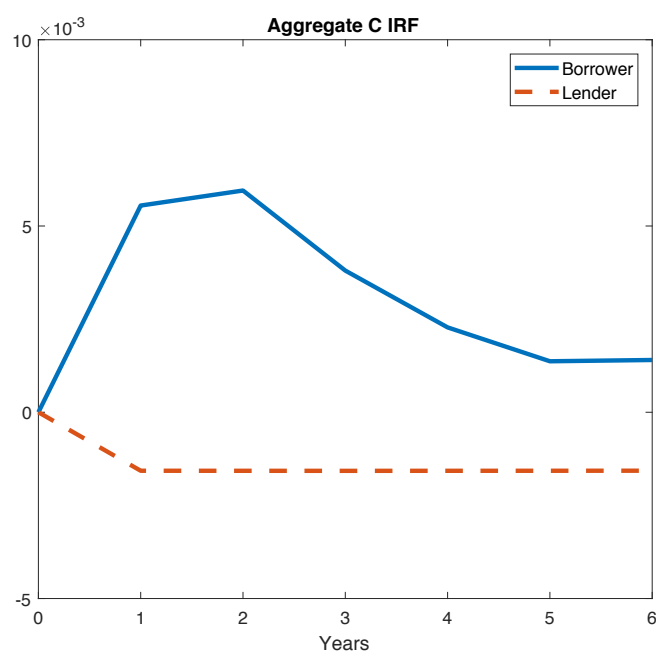


Figure shows the separate contributions of borrowers and lenders to the aggregate consumption impulse response in the 2008 baseline model shown in Figure XI.

Figure A-13: Effect of σ on $\Delta IRF_{f_0(\mu, \sigma)}$

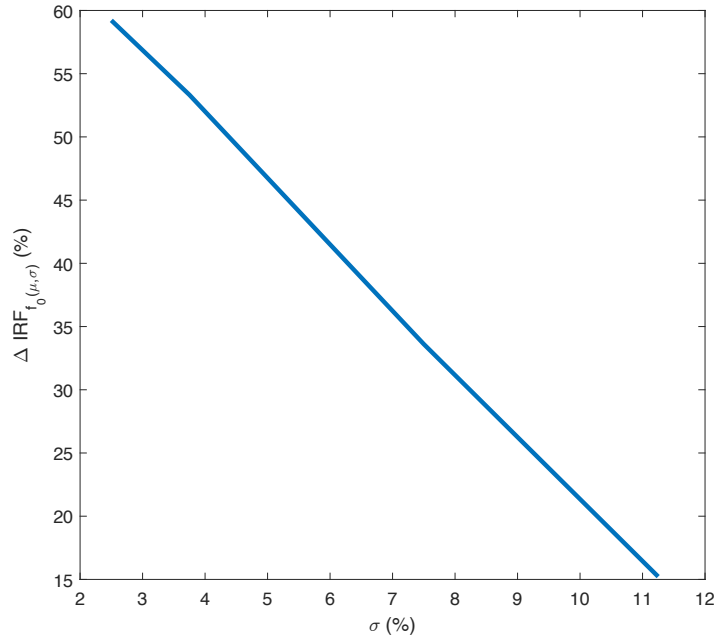
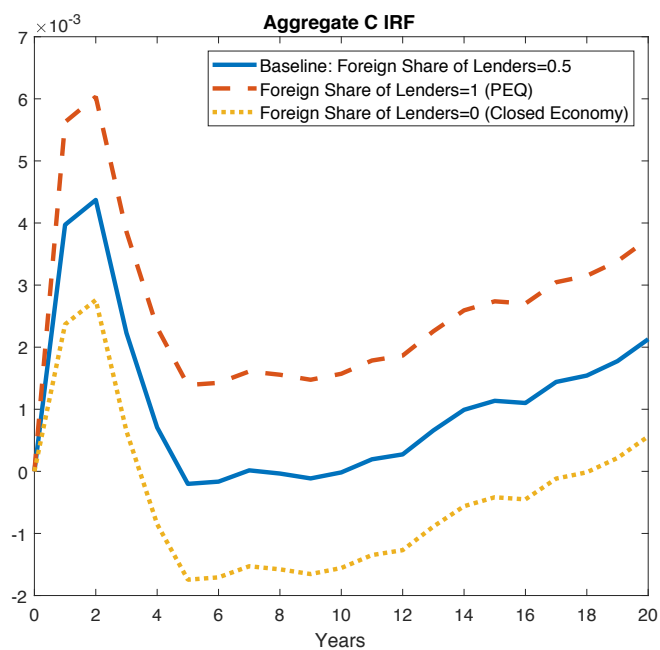


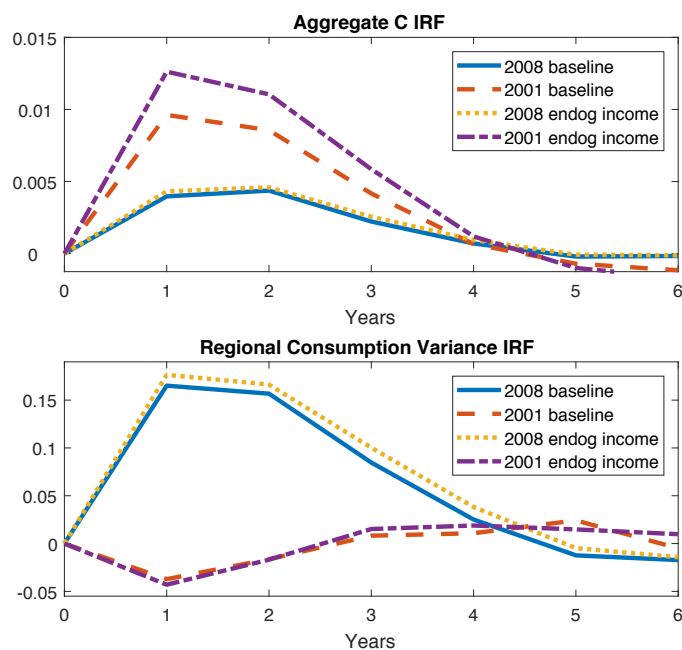
Figure shows how much a 10% increase in equity changes the aggregate consumption response to a 1% decrease in r^m , for different values of the cross-region standard deviation of equity. Let $IRF_{f_0(\mu, \sigma)} = \frac{\Delta C}{\Delta r^m}$ be the response on impact (in the first year) of consumption to a 1% decline in interest rates given an initial equity distribution f_0 with mean μ and standard deviation σ . Then let $\Delta IRF_{f_0(\mu, \sigma)} = \frac{IRF_{f_0(\mu+10\sigma)} - IRF_{f_0(\mu, \sigma)}}{IRF_{f_0(\mu, \sigma)}}$. $\Delta IRF_{f_0(\mu, \sigma)}$ measures how a 10 percentage point increase in the mean of the equity distribution affects the impulse response of consumption to interest rates. This figure plots $\Delta IRF_{f_0(\mu, \sigma)}$ as we change the cross-region standard deviation of equity σ . The leftmost point in the figure sets $f_0(\mu, \sigma)$ equal to the stochastic steady-state distribution, and then we steadily increase the cross-region standard deviation of equity. In the stochastic steady-state with $\sigma = 2.5\%$, a 10% increase in house prices increases the aggregate consumption response to a 1% rate decline by almost 60%. However, when σ is higher, this same mean increase in house prices has a smaller effect on the response of consumption to a rate drop.

Figure A-14: Importance of GE Effects from Lenders' Consumption



This figure shows aggregate consumption IRFs for alternative assumptions on what share of mortgage debt is held by domestic households, whose income will fall when interest rates decline.

Figure A-15: Endogenous Income Effects



This figure shows outcomes for a version of the model which partially endogenizes income.

Figure A-16: Adjustable-Rate Mortgage Shares pre-QE1 and Payment Reductions over Nov 2008 – June 2009

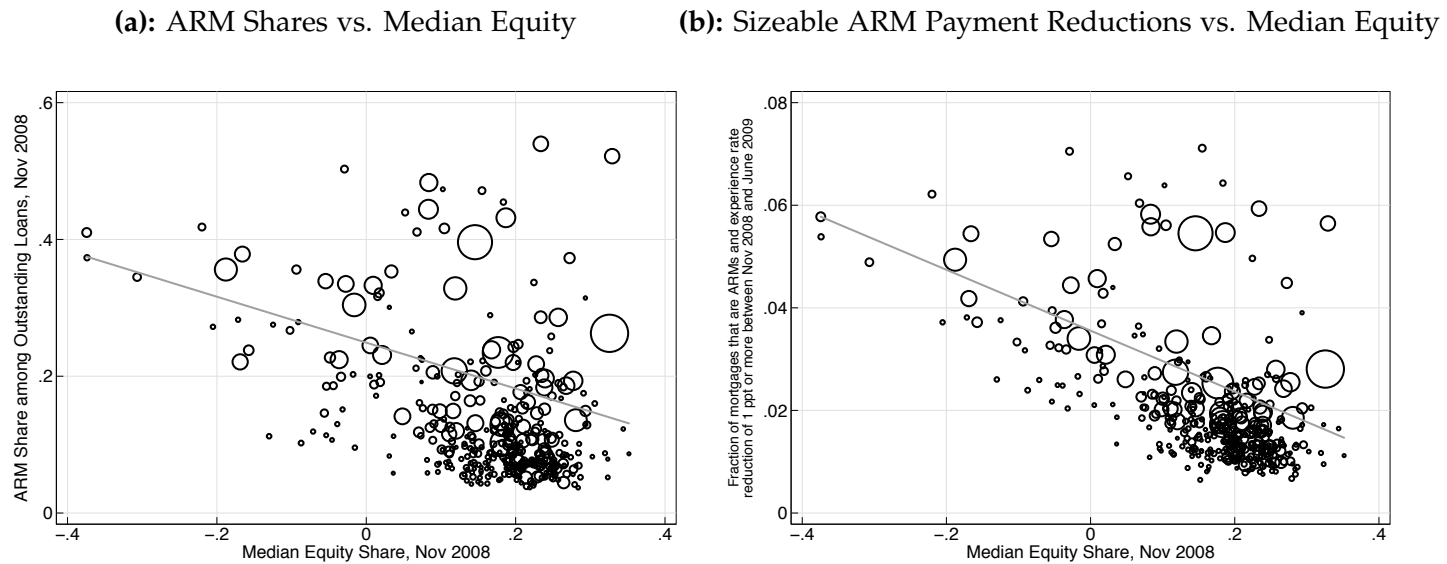
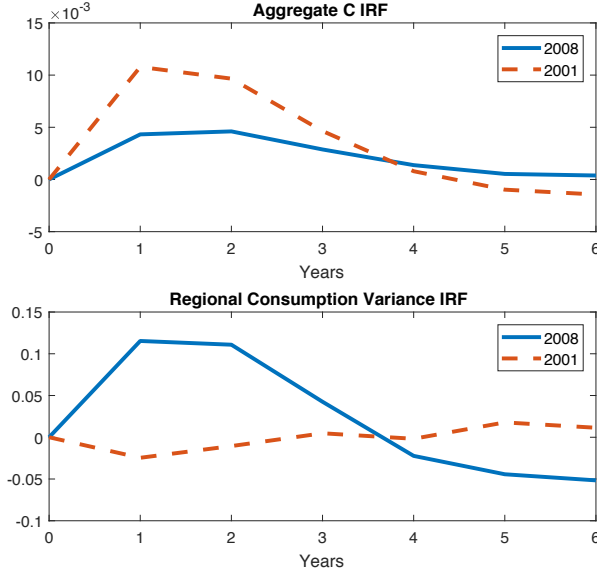


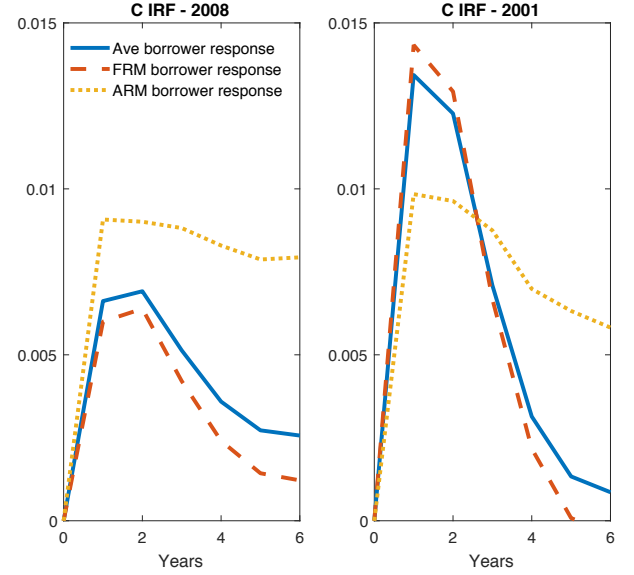
Figure shows scatter plots of the balance-weighted fraction of adjustable-rate mortgages (ARMs) in a given MSA (panel a) or the balance-weighted fraction that are ARMs and experience a rate reduction of 1 percentage point or more over November 2008 – June 2009 (panel b), as measured in the CRISM data, versus the median equity share (E^{med}) within the MSA as of November 2008. When computing the share with rate drops, we also require that the recorded required monthly payment not increase over the same period (which may indicate that the loan's IO period expired). The size of the circle represents the 2008 population of the MSA. The gray lines represent simple regression lines fitting the scatter plot (population weighted). The regression line in panel (b) has a slope of -0.059 (s.e. = 0.009). Meanwhile, regressing the 6-month refinancing propensity (from January-June 2009) on $E_{j,Nov2008}^{med}$ yields a coefficient of +0.147 (s.e. = 0.028).

Figure A-17: Accounting for Ex-Ante Heterogeneity

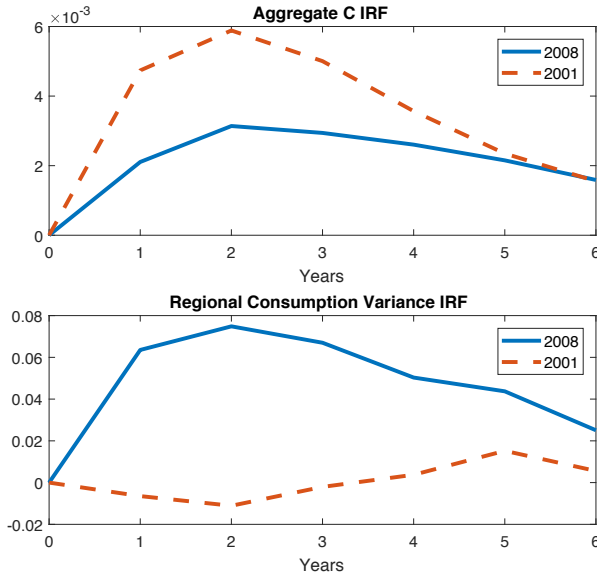
(a): Match ARM-Share



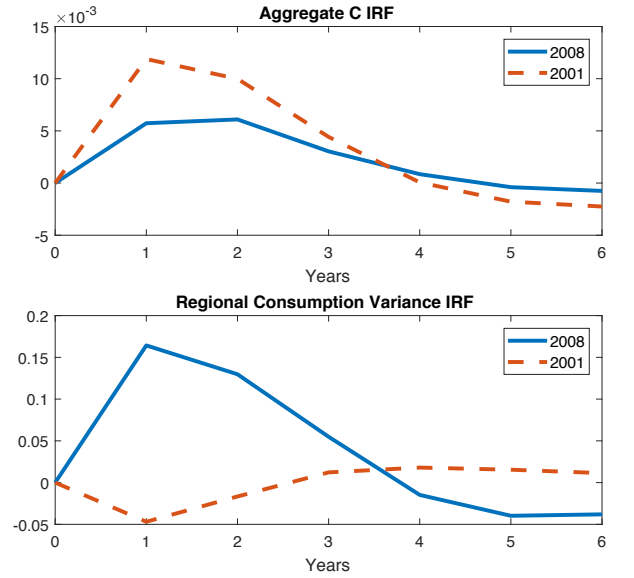
(b): Spending responses of ARM vs. FRM borrowers



(c): Reduce Equity-Refi Elasticity

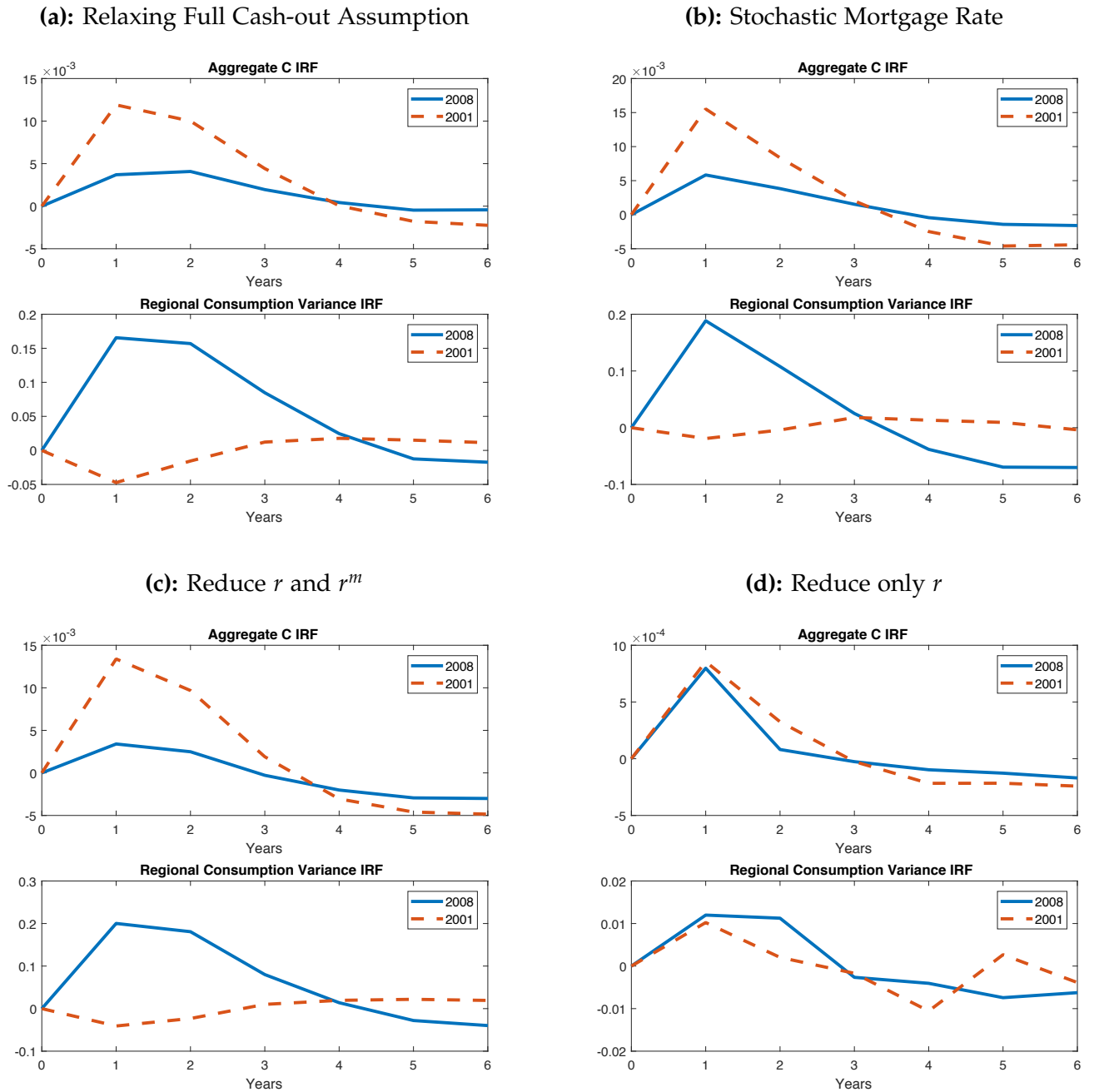


(d): Boom-Bust



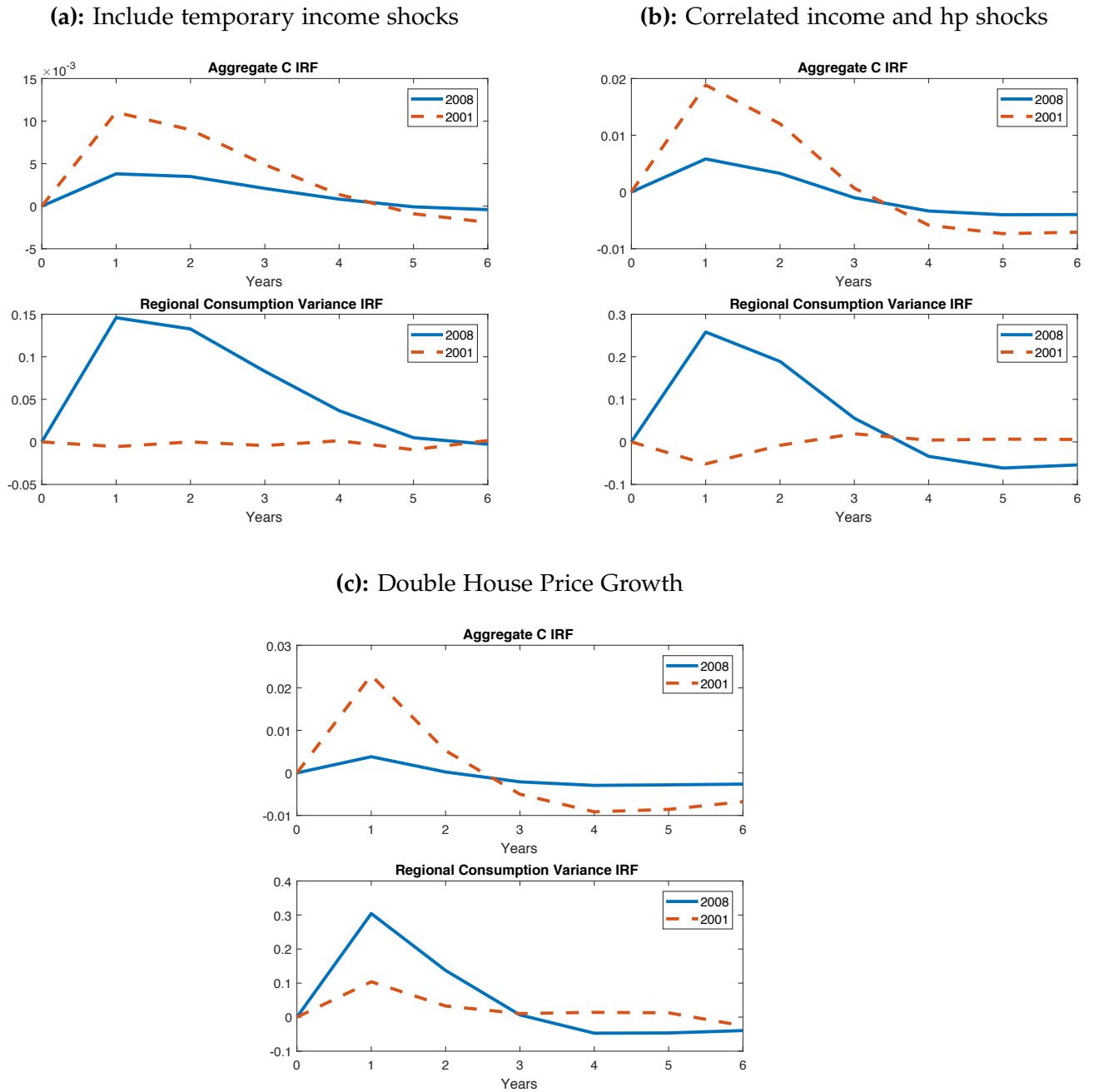
Panel (a) repeats Figure XI in a model with both FRM and ARMs calibrated to match the regional differences with house prices in the data. Panel (b) shows the average spending response of ARM borrowers, FRM borrowers and their share-weighted averages in 2008 compared to 2001. Note that the share-weighted average borrower average is not identical to the aggregate spending response since it includes no lender responses. Panel (c) repeats Figure XI in a model where fixed costs are doubled. This roughly halves the relationship between regional equity and refinancing. Panel (d) repeats Figure XI in a model where regions with the largest house price declines experience a prior house price boom.

Figure A-18: Robustness Results



This figure repeats Figure XI in several alternative models. In panel (a) households can choose between full cash-out and no cash-out refinancing. In panel (b) interest rate movements are stochastic instead of a one-time shock. In panel (c) r and r^m both decline, with a constant spread. In panel (d), only r declines.

Figure A-19: More Robustness Results



This figure repeats Figure XI in several alternative models. In panel (a) households experience i.i.d. transitory income shocks. In panel (b), regional house price and income shocks are perfectly correlated instead of uncorrelated. In panel (c) trend house price growth is doubled.